

Software Requirement Specifications (SRS)
Software Design Document (SDD)
INVENTORY MANAGEMENT SYSTEM
(SMART STOCK)



Submitted By

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Software Requirement Specifications

Table of Contents

1. Introduction	1
1.1 Objectives	1
1.2 Project boundaries	1
1.3 References	1
1.4 Overview	1
1.5 Definitions, Acronyms, and Abbreviations	2
2. The Overall Description	3
2.1 System Context:	3
2.1.1 Main advantages:	3
2.1.2 Daily Activities:	3
2.1.3 Deployment Needs:	3
2.1.4 Core Features:	3
2.2 User Profiles:	4
2.2.1 Owner / Administrator	4
2.2.2 Employee / Cashier	4
2.2.3 Assumptions	4
3. Detail Requirement	4
3.1 User Interface Requirements	4
3.1.1 Owner Panel	4
3.1.2 Employee Panel:	5
3.1.3 Hardware Needs:	5
3.1.4 Development Environment:	5
3.2 Functional Requirements:	5
3.3 Use Cases (How user will interact with the system) :	6
3.4 Classes / Objects:	7

3.5 Data Flow Diagrams (DFDs):	8
3.5.1 Level 0 Data Flow Diagram	8
3.5.2 Level 1 Data Flow Diagram	9
2.5.3 Level 2 Data Flow Diagram	10
3.6 Non-Functional Requirements	11
3.7 Logical Database Requirements	12
3.8 Design Constraints	12
3.8.1 Standards Compliance	12

1. Introduction

1.1 Objectives

The Inventory Management System is a modern web platform created to simplify and improve the control over stock levels, customer sales, and supplier purchases in retail outlets. Its core aim is to replace paper-based methods, eliminate mistakes, and raise daily efficiency through smart automation. This solution is built especially for small and medium retail shops to help them run smoother operations and achieve stronger results.

1.2 Project boundaries

This browser-based application offers one central place to handle stock monitoring, transaction recording, and procurement activities. The coverage includes

- Unified dashboard for all store activities.
- Protected entry according to user roles.
- Live statistics and instant report creation.
- Designed mainly for small to medium businesses.

1.3 References

<https://www.slideshare.net/>

www.google.com

www.w3schools.com

1.4 Overview

This specification explains all functional and non-functional needs, architecture guidelines, and development roadmap for the Inventory Management System so that developers and stakeholders share the same vision.

Key capabilities:

- Safe login and session control.
- Full product and stock oversight.
- Sales and buying workflow support.
- Detailed analytics and printable summaries.

Focus is placed on simplicity, growth potential, and user comfort.

1.5 Definitions, Acronyms, and Abbreviations

Abbreviation	Definition
SRS	Software Requirements Specification: Software Requirements Specification (SRS) refers to the formal document that defines what the software under development should do, covering its functional behavior, performance, performance, security, and safety aspects
DFD	A Data Flow Diagram (DFD) is a graphical representation that illustrates how data moves through a system.
WCAG	WCAG stands for Web Content Accessibility Guidelines. It is a set of technical standards created by the W3C to make web content more accessible to users with various disabilities.
RAM	Random Access Memory (RAM) is high-speed volatile memory used by the computer to store data and instructions required for immediate processing and efficient operation.
SQL	SQL (Structured Query Language) is a domain-specific language used for creating, reading, updating, and deleting data in relational databases.
ISO	International Organization for Standardization
POS	Point of Sale : It denotes the moment and place at which a purchase or sale transaction is completed at the retail counter.
PHP	PHP: Hypertext Preprocessor: It is a popular programming language created with web development in mind.
XAMPP	It refers to a free, open-source, cross-platform package known as a web server solution stack, provided by Apache Friends, which consists of the Apache HTTP Server and MariaDB database.

2. The Overall Description

The system works as an all-in-one online tool that automates and upgrades routine retail tasks covering stock handling, checkout processes, ordering from suppliers, and performance reviews.

2.1 System Context:

It is a self-contained web solution ideal for small and medium stores, combining stock control, sales tracking, purchase records, and reporting without needing extra programs.

2.1.1 Main advantages:

- Works in any updated browser.
- Single reliable database with immediate updates.
- Ready to grow together with the business.

2.1.2 Daily Activities:

- Recording customer payments and printing bills while reducing stock instantly.
- Checking stock regularly, receiving low-stock warnings, and planning reorders.
- Helping buyers, managing returns or swaps.
- Every employee must log in securely; administrators watch user actions.

2.1.3 Deployment Needs:

- Only administrators may change core settings.
- Works perfectly on Chrome, Firefox, and Edge.
- Clear screens designed for each role.
- Automatic nightly backups managed by admin.

2.1.4 Core Features:

- Stock Handling.
- Live quantity tracking and automatic alerts when items run low.
- Create, modify, or remove products.
- Group items into categories.
- Checkout Module.
- Bill storage and sales history.
- Procurement Module.
- Raise orders to vendors and follow their status.
- Update stock automatically when goods arrive.
- Analytics Section.
- Reports on revenue, stock status, and spending
- Export to PDF or Excel

- Access & Security
- Different permission levels (Owner / Employee)
- Strong password system (future two-factor option)

2.2 User Profiles:

2.2.1 Owner / Administrator

- Adjust store information, taxes, etc.
- Add or remove staff accounts and set their rights
- Full control over products, sales history, and reports
- Manage backups and security rules

2.2.2 Employee / Cashier

- Ring up sales and print receipts
- Check current stock during customer service
- Handle approved returns
- Restricted menu according to store policy
- 2.4 Limitations
- Needs constant internet connection
- Best viewed on desktop screens (mobile view limited in first version)
- No direct link to external accounting packages at launch

2.2.3 Assumptions

- Reliable internet at the store
- Users have modern computers and browsers
- Basic equipment present (PC, barcode reader, receipt printer)

3. Detail Requirement

3.1 User Interface Requirements

3.1.1 Owner Panel

- Staff account and permission control
- Product catalogue maintenance
- Advanced reporting and settings area

3.1.2 Employee Panel:

- Quick POS screen with barcode scanning
- Instant receipt printing
- Fast product search

3.1.3 Hardware Needs:

- CPU 1.6 GHz or better
- 4 GB RAM minimum
- Internet connection
- At least 350 MB free disk space

3.1.4 Development Environment:

- Editor: Visual Studio Code
- Server: XAMPP (Apache + PHP + MySQL)

3.2 Functional Requirements:

- Full create/update/delete for items and stock
- Automatic stock change after every sale or receipt
- Invoice creation and permanent storage
- Supplier records and purchase order tracking
- Secure role-based login
- Flexible reports on demand

3.3 Use Cases (How user will interact with the system) :

Use cases capture the step-by-step interaction between users and the system to achieve specific business goals while validating functional requirements.

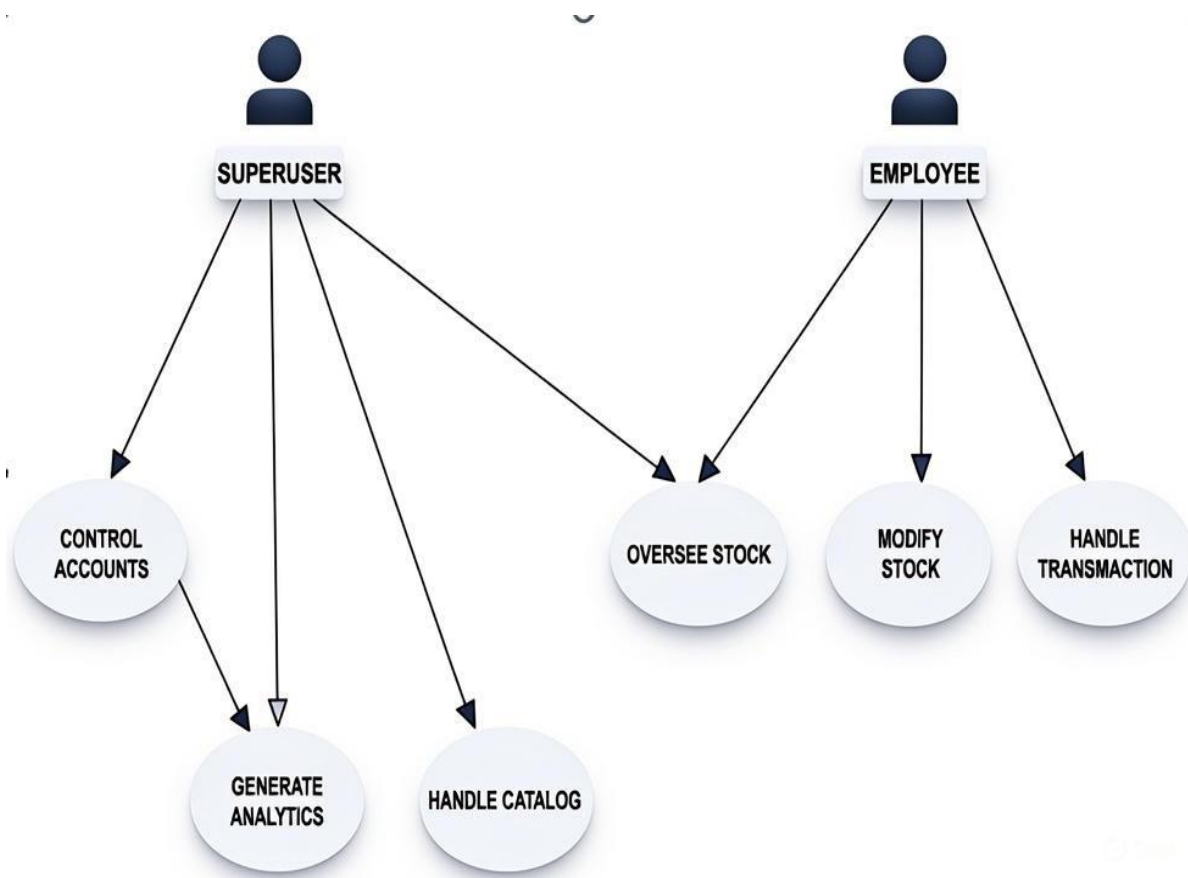


Figure 3.3: Use Cases

3.4 Classes / Objects:

This section contains major classes of the *INVENTORY MANAGEMENT SYSYTEM*.

A class in a class diagram is usually represented as a rectangle divided into three sections:

- **Top Section:** Class Name
- **Middle Section:** Attributes
- **Bottom Section:** Methods

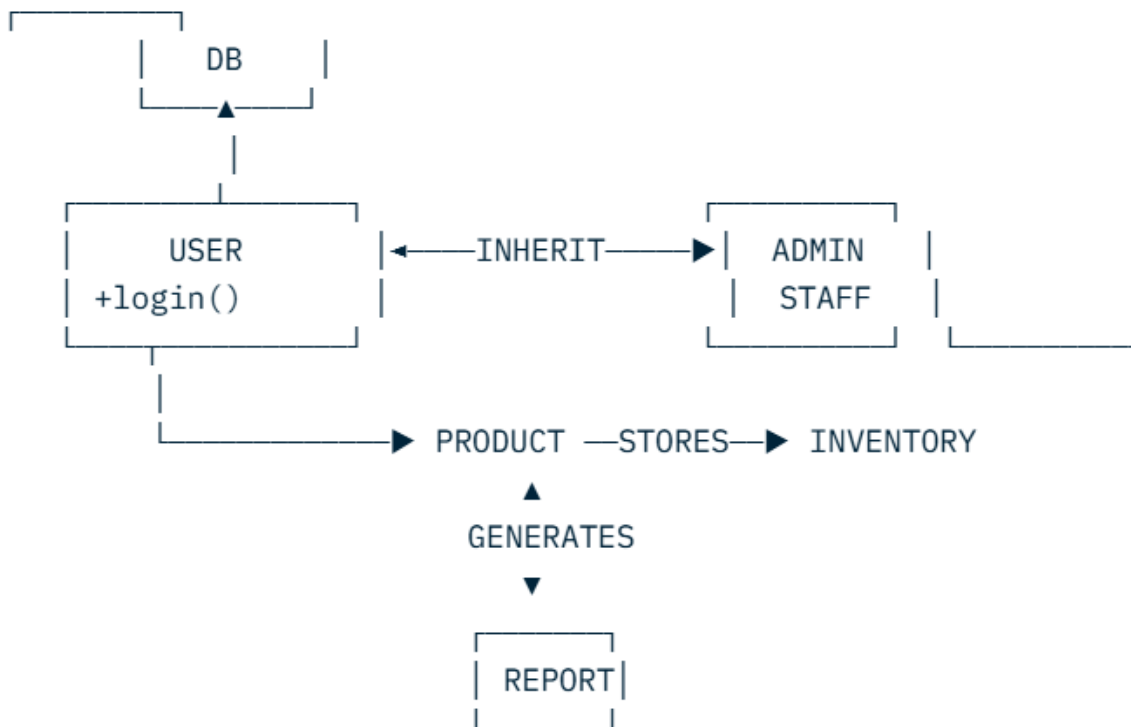


Figure 3.4.1: Class Diagram

3.5 Data Flow Diagrams (DFDs):

A Data Flow Diagram is a visual representation of how data moves through a system. It enhance understanding and help in identifying new issues while creating a new system.

3.5.1 Level 0 Data Flow Diagram

The **Level-0 DFD**, also referred to as the Context Diagram, presents an overall picture of the system. It depicts the system as a single central process and highlights the interactions and data exchanges with external entities (actors), while omitting internal details and data stores.

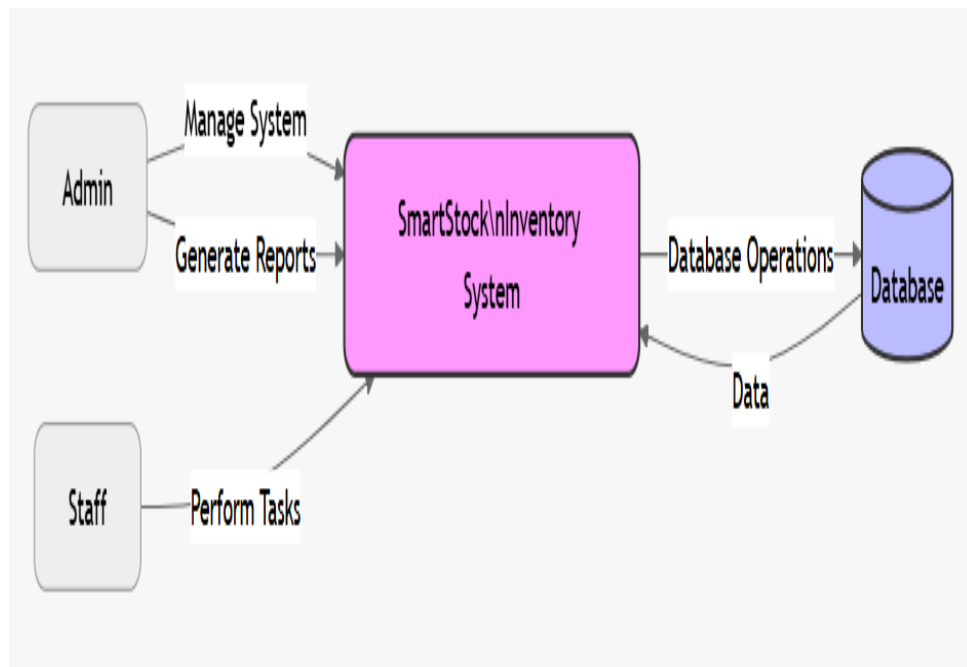


Fig 3.5.1: 0 Level DFD

3.5.2 Level 1 Data Flow Diagram

A **Level-1 Data Flow Diagram** expands the high-level process of the Level-0 DFD by breaking it down into primary subprocesses. It shows detailed internal data flows and emphasizes the system's inner functionality.

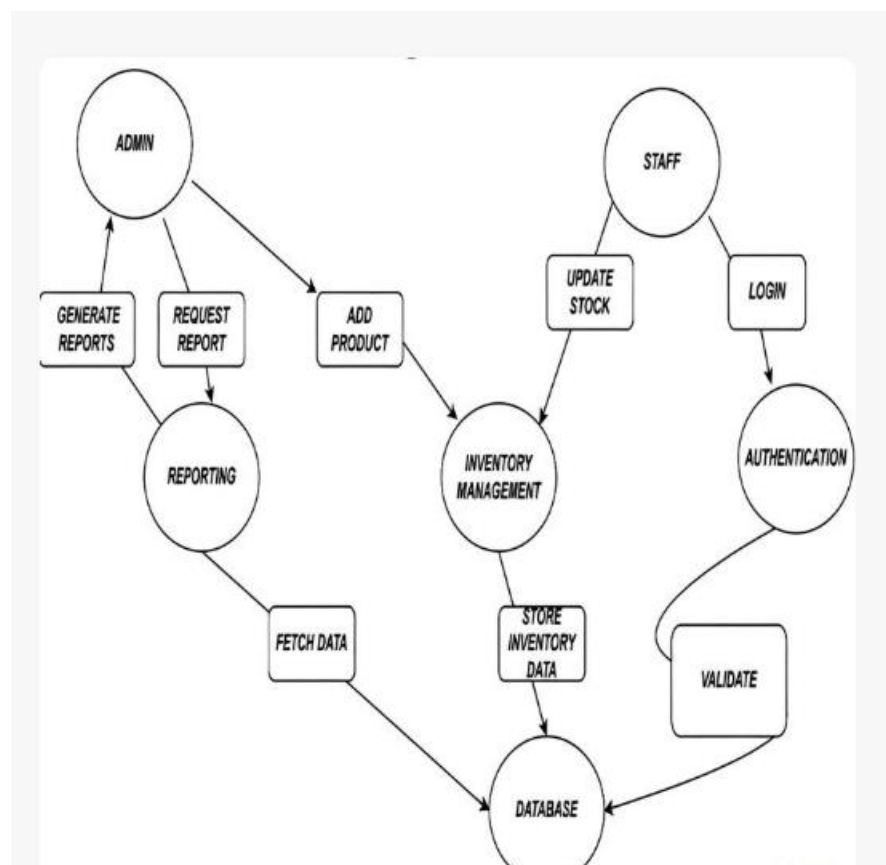


Fig 3.5.2: 1 Level DFD

2.5.3 Level 2 Data Flow Diagram

The **Level-2 DFD** zooms into a single subprocess identified in the Level-1 diagram, revealing its inner workings. It illustrates the finer-grained operations and data flows inside that subprocess and can include extra data stores and interactions.

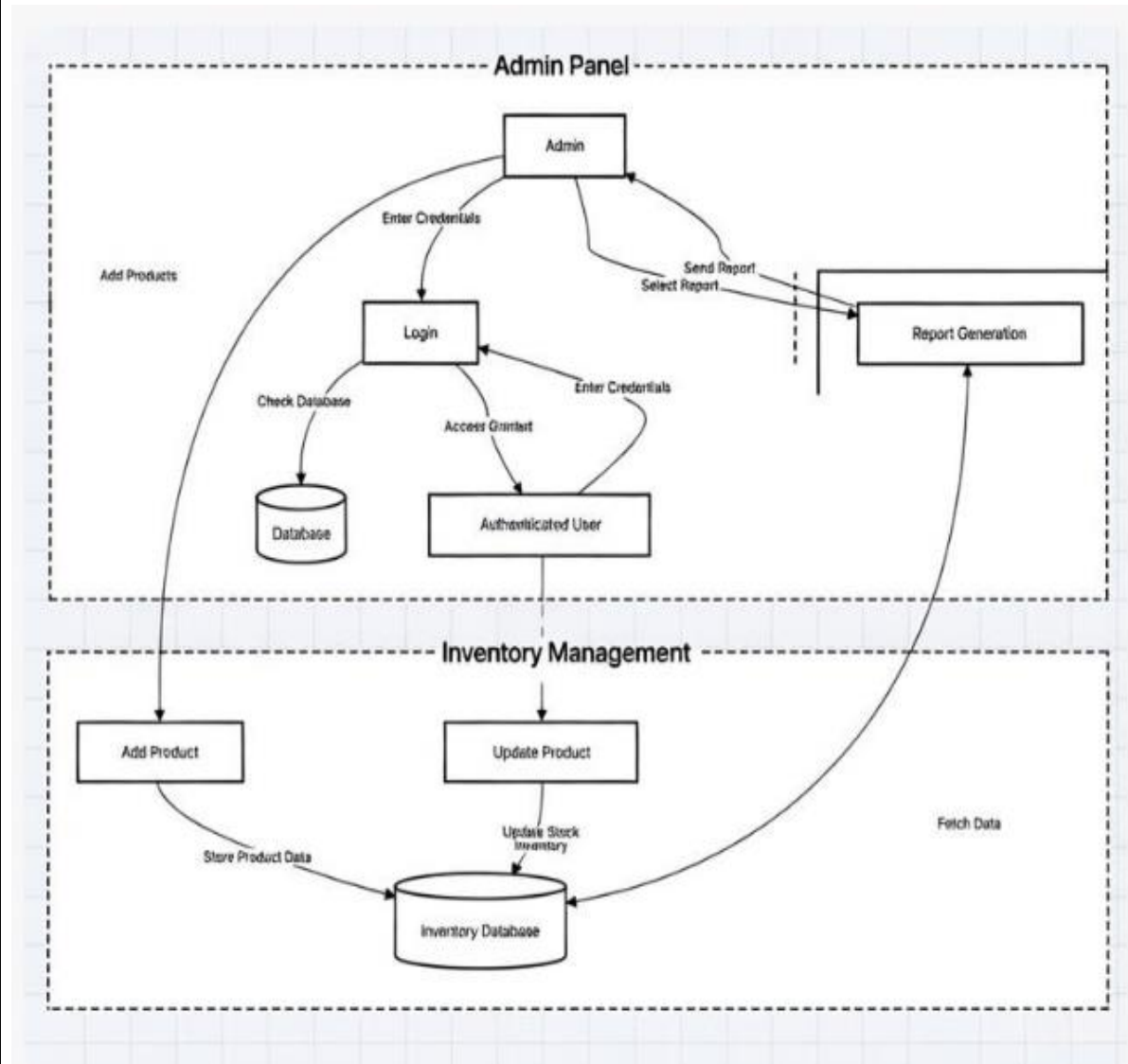


Fig 3.5.3: 2 Level DFD

3.6 Non-Functional Requirements

Non-Functional Requirements

- **Usability:** The system shall provide an intuitive and user-friendly interface that enables users of all experience levels to perform tasks efficiently with minimal training.
- **Performance:** The system shall deliver response times of less than 5 seconds for all user interactions and critical operations, even under peak transaction loads.
- **Scalability:** The system shall support seamless horizontal and vertical scaling to accommodate growing numbers of users, data volume, and transaction throughput without degradation in performance.
- **Security:** Sensitive data shall be protected through end-to-end encryption, role-based access control (RBAC), secure authentication mechanisms, and full compliance with applicable data protection regulations (e.g., GDPR, CCPA).
- **Reliability:** The system shall achieve at least 99.9% availability (measured monthly), with automated daily backups and a documented disaster recovery plan ensuring data restoration within 4 hours in case of major incidents.
- **Maintainability:** The system shall be designed with modular, well-documented, and loosely coupled components to facilitate future enhancements, bug fixes, and technology upgrades with minimal effort.
- **Compliance:** The system shall fully comply with all relevant legal and regulatory requirements, including financial, tax, invoicing, and industry-specific standards applicable to the business jurisdiction(s).
- **Portability:** The system shall be fully functional and consistently usable across major platforms and devices, including desktop browsers (Windows/macOS/Linux), tablets, and mobile devices (iOS and Android).
- **Accessibility:** The system shall conform to WCAG 2.1 Level AA standards, ensuring usability for people with disabilities, including support for screen readers, keyboard navigation, and sufficient color contrast.
- **Backup and Recovery:** The system shall perform automated, encrypted daily backups with point-in-time recovery capabilities and maintain a tested disaster recovery procedure to restore full operations within defined recovery time objectives (RTO) and recovery point objectives (RPO).

3.7 Logical Database Requirements

The inventory management system demands a robust and well-structured database to guarantee data accuracy, protection, high performance, and future growth while fully supporting day-to-day business activities.

Key database design principles applied:

- **Architecture:** Fully relational design relying on primary and foreign key constraints to maintain referential integrity.
- **Core objects:** Covers all essential areas — staff accounts, product catalog, sales records, stock levels, vendor details, and analytical reporting.
- **Field types:** Carefully chosen data types (e.g., character strings via VARCHAR, whole numbers via INT, dates via DATE/DATETIME, monetary values via DECIMAL) to optimize storage and speed.
- **Normalization level:** Schema is normalized to at least Third Normal Form (3NF), eliminating redundancy and preventing update anomalies.

3.8 Design Constraints

3.8.1 Standards Compliance

Regulatory and Industry Compliance Obligations

- **Card Payment Security** Every component involved in accepting, processing, or storing payment card information shall fully adhere to the latest Payment Card Industry Data Security Standard (PCI DSS) requirements.
- **Management System Certifications** The overall development, operational, and quality assurance processes must follow the guidelines of ISO 9001 (Quality Management) and ISO/IEC 27001 (Information Security Management System).
- **Personal Data Protection** The platform shall remain fully compliant with the General Data Protection Regulation (GDPR), providing robust safeguards for personal data, transparent processing notices, and full support for data-subject rights including access, correction, deletion, restriction, portability, and objection.
- **Inclusive Design & Accessibility** All user-facing interfaces (web portals, mobile apps, and printed/digital receipts) must satisfy Web Content Accessibility Guidelines (WCAG) 2.1 Level AA success criteria to guarantee usability for individuals with visual, auditory, motor, or cognitive impairments.
- **Third-Party Integration Governance** Connections to external systems—payment providers, shipping carriers, accounting tools, or any other service—must use industry-standard, well-documented APIs, enforce mutual TLS encryption, implement OAuth 2.0 or equivalent authentication, include proper timeout/retry logic, and respect rate limits.
- **Taxation & Fiscal Compliance** The solution must automatically apply correct local tax rates, produce legally valid invoices and credit notes, maintain immutable audit trails for all financial

transactions, and generate statutory sales and tax reports in the exact format required by regional revenue authorities.

- **Secure Development Lifecycle** All code must be written according to recognized secure coding standards (e.g., OWASP Top Ten mitigation techniques, CERT secure coding rules), include comprehensive inline documentation, undergo regular static and dynamic security testing, and follow version-controlled, peer-reviewed release processes to ensure long-term maintainability and security.

Software design Document

Revision History

Date	Detail	Writer	Remarks

Approval Page

This design document has been reviewed and formally approved by the following authorities:

Signature	Name	Title	Date

Software design Document

Table of Contents

1. Introduction 1

1.1 Aim	1
1.2 Project Coverage	1
1.3 General Description.....	1
1.4 Source	2
1.5 Definitions, Acronyms, and Abbreviations.....	2

2. System Summary.....3

3. SYSTEM ARCHITECTURE4

3.1 Design Sutructure	4
3.2 Decomposition Description.....	7
3.2.1 Data Flow Diagram	7
3.2.2 ER Diagram.....	9
3.2.3 Activity Diagram.....	10
3.2.3 Sequence Diagram	11

4. DATA DESIGN 13

5. USER INTERFACE DESIGN 14

5.1 Overview of User Interface.....	14
5.2 Screen Images	14
5.2.1 Home Screen.....	14
5.2.2 Login Screen	15

6. Appendices 15

6.1 References	15
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1. Introduction

1.1 Aim

This document (SDD) explains the technical layout of the Inventory Management System. It provides a clear picture of how the system will be arranged, built, and connected. It is written for programmers, reviewers, and project members so they can easily understand how the system will be developed. It also describes design decisions, system parts, and how different sections will work together based on the requirement document.

1.2 Project Coverage

The Inventory Management System is an online solution developed to support small and medium shops in handling their daily work. It manages stock levels, sales records, purchase details, reports, and user access. By changing manual work into automatic processes, it saves time, lowers mistakes, and helps shop owners make better choices.

Main Capabilities:

- Continuous stock monitoring with warning alerts.
- Sales handling with bill creation and payment tracking.
- Supplier and purchase control.
- Secure login with user roles.
- Reports for business planning.

The system works through internet browsers and is designed to grow as business needs increase. Future versions may include mobile access and external connections.

1.3 General Description

This system offers a single platform for managing shop operations. Automation and central control reduce workload and increase accuracy. Managers and workers can use it easily through a clean and simple interface.

Targets:

- Provide a reliable and expandable system.
- Reduce paperwork through automation.
- Give useful business insights.
- Improve ease of use with role base access.

The system design follows modern standards of performance, safety, and accessibility. Its modular style allows easy updates later.

1.4 Sources

The following materials supported the system design:

- Online slides about store systems.
- Internet articles about web development.
- Online encyclopaedias for technical ideas.
- Learning sites for HTML, CSS, and PHP.
- Accessibility guidelines.
- International quality stander. □

1.5 Definitions, Acronyms, and Abbreviations

Acronym	Full Form	Definition
SRS	Software Requirements Specification	Requirement description document.
ER Diagram	Entity-Relationship Diagram	Database structure drawing.
DFD	Data Flow Diagram	Data movement chart.
POS	Point of Sale	Sales point system.
UI	User Interface	User interaction area.

RBAC	Role-Based Access Control	Access by role method.
WCAG	Web Content Accessibility Guidelines	Accessibility rules.
PHP	Hypertext Preprocessor	A server-side scripting language used for web development.
MDN	Mozilla Developer Network	An online resource providing documentation for web technologies.
GDPR	General Data Protection Regulation	Data privacy law.
IEEE	Institute of Electrical and Electronics Engineers	Engineering standards group.
PCI DSS	Payment Card Industry Data Security Standard	Payment safety rules.
HTML	Hypertext Markup Language	A markup language used for creating web pages.
CSS	Cascading Style Sheets	Page design language.
MySQL	My Structured Query Language	Database software.
XAMPP	Cross-Platform, Apache, MariaDB, PHP, and Perl	Local server tool.

2. System Summary

The Inventory Management System is a flexible web-based platform that joins stock, sales, purchases, and reports in one place. It supports smooth daily operations for shop staff and owners.

Main Sections:

Stock Control:

- Livestock updates.
- Low-stock alerts.
- Product Grouping.

Sales Control:

- Fast selling process.
- Bill and payment records.
- Sales performance reports.

Purchase Control:

- Supplier data handling.
- Order and restock tracking.

Reports:

- Sales and stock summaries.
- Planning support data.

Users:

- Different permission levels.
- Protected login system.

3. SYSTEM ARCHITECTURE

3.1 Design Structure

The system follows a layered design to improve speed, safety, and maintenance.

Layers:

Interface Layer:

- Screens for admin and staff.
- Works on modern browsers.
- Ready for mobile later.

Logic Layer:

- Controls system rules.
- Built with PHP and JavaScript.

Storage Layer:

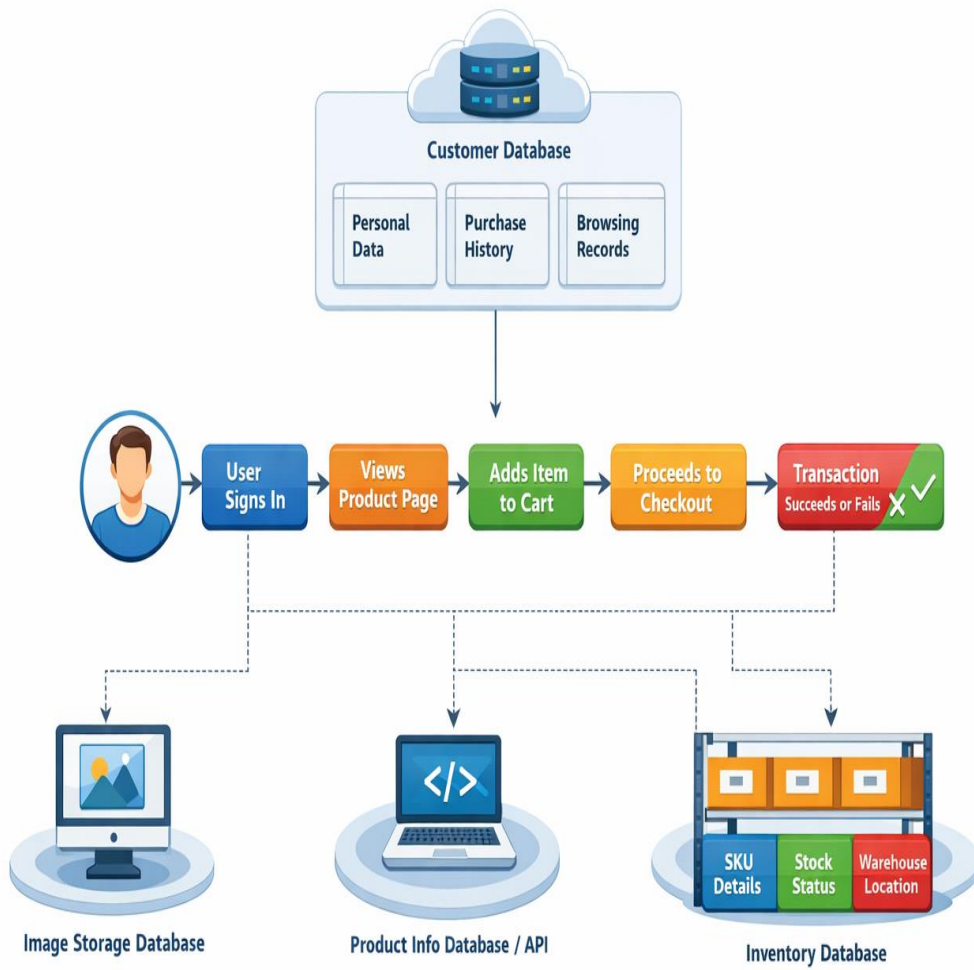
- MySQL database.
- Organized tables for fast access.

Deployment:

- Runs on Apache server.
- Backup and recovery support.

Overall Flow:

User → Web Server → Database



3.2 Decomposition Description

3.2.1 Data Flow Diagram

A Data Flow Diagram (DFD) is a visual tool that shows how data moves through a system. It illustrates the transformation of input data into output data through various processes, while also depicting the interactions between processes, data stores, and external entities (such as users or other systems).

3.2.1.1 Context level Data Flow Diagram (DFD)

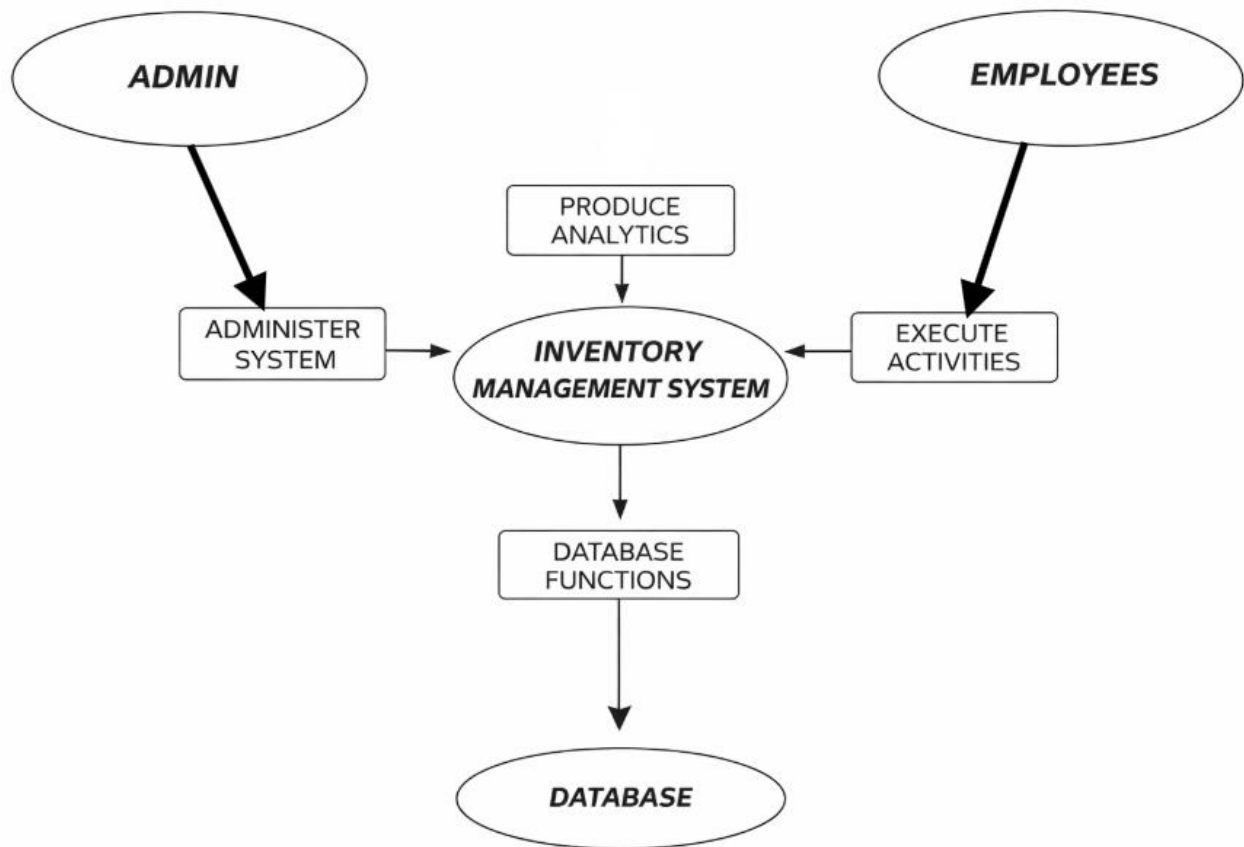


Figure 3.2.1.1: 0 Level Diagram

3.2.1.2 Level 1 Data Flow Diagram (DFD)

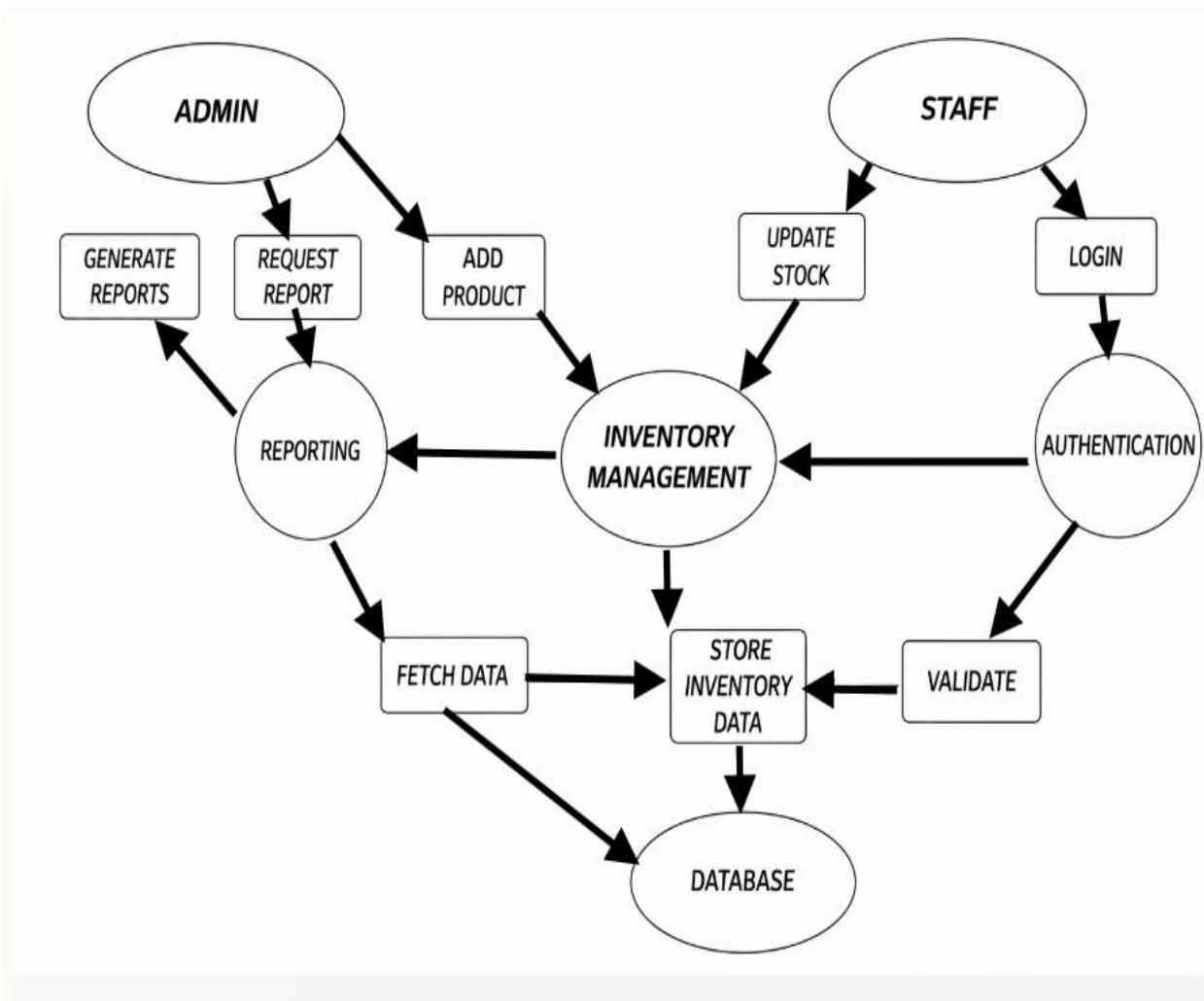


Figure 3.2.1.1: 1 Level Diagram

3.2.2 ER Diagram

An Entity-Relationship (ER) Diagram is a conceptual modeling tool that graphically represents the entities relevant to a system, along with the relationships that exist among them. It serves as a foundational step in database design, providing a clear blueprint that can be systematically converted into a relational database schema.

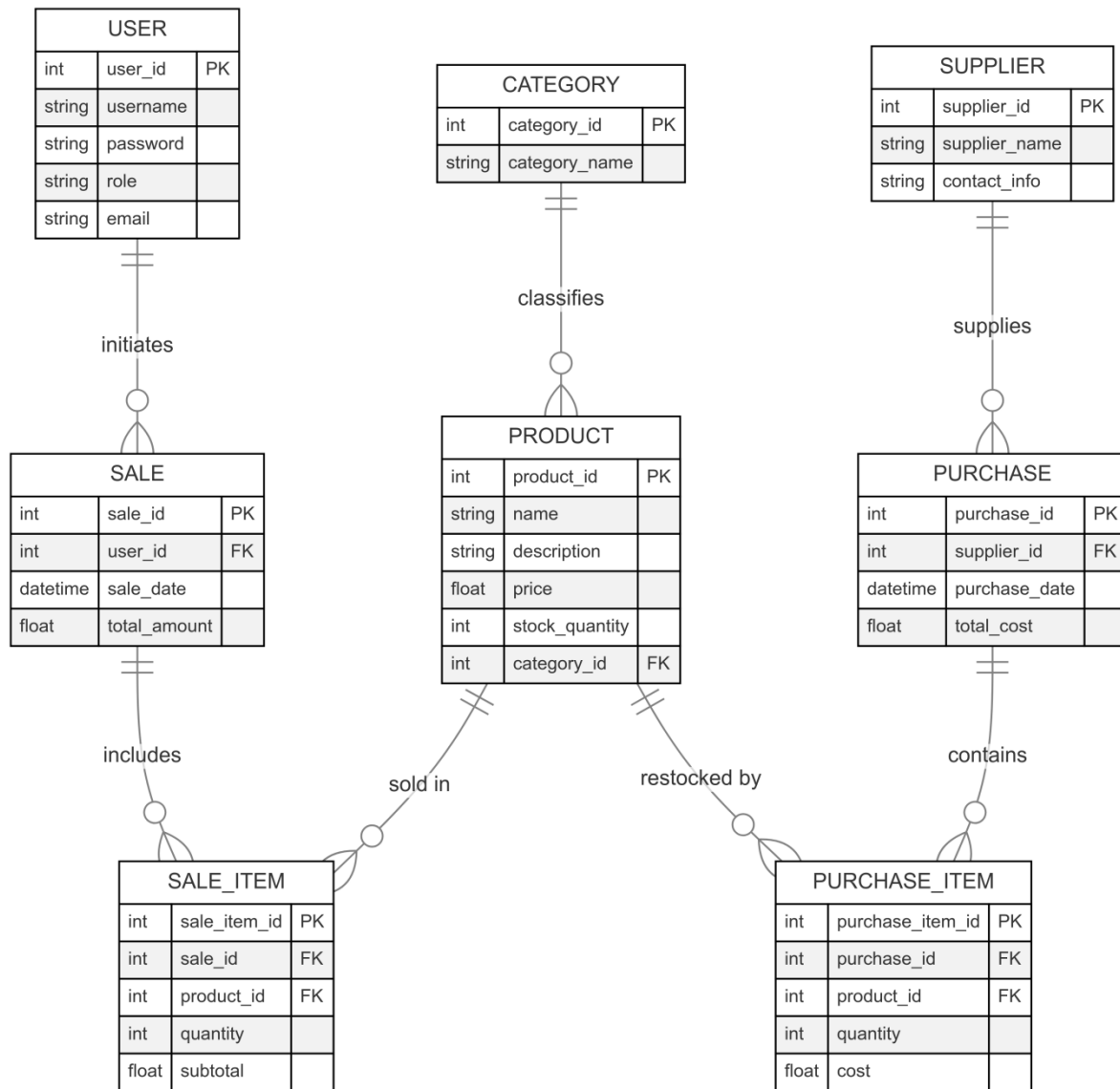


Figure 3.2.2: ER Diagram

3.2.3 Activity Diagram

An Activity Diagram is a UML diagram that visually shows the step-by-step flow of activities or tasks in a system or business process. It illustrates the sequence of actions, decision points, branching paths, and parallel (concurrent) activities, making it easy to understand how the system operates over time.

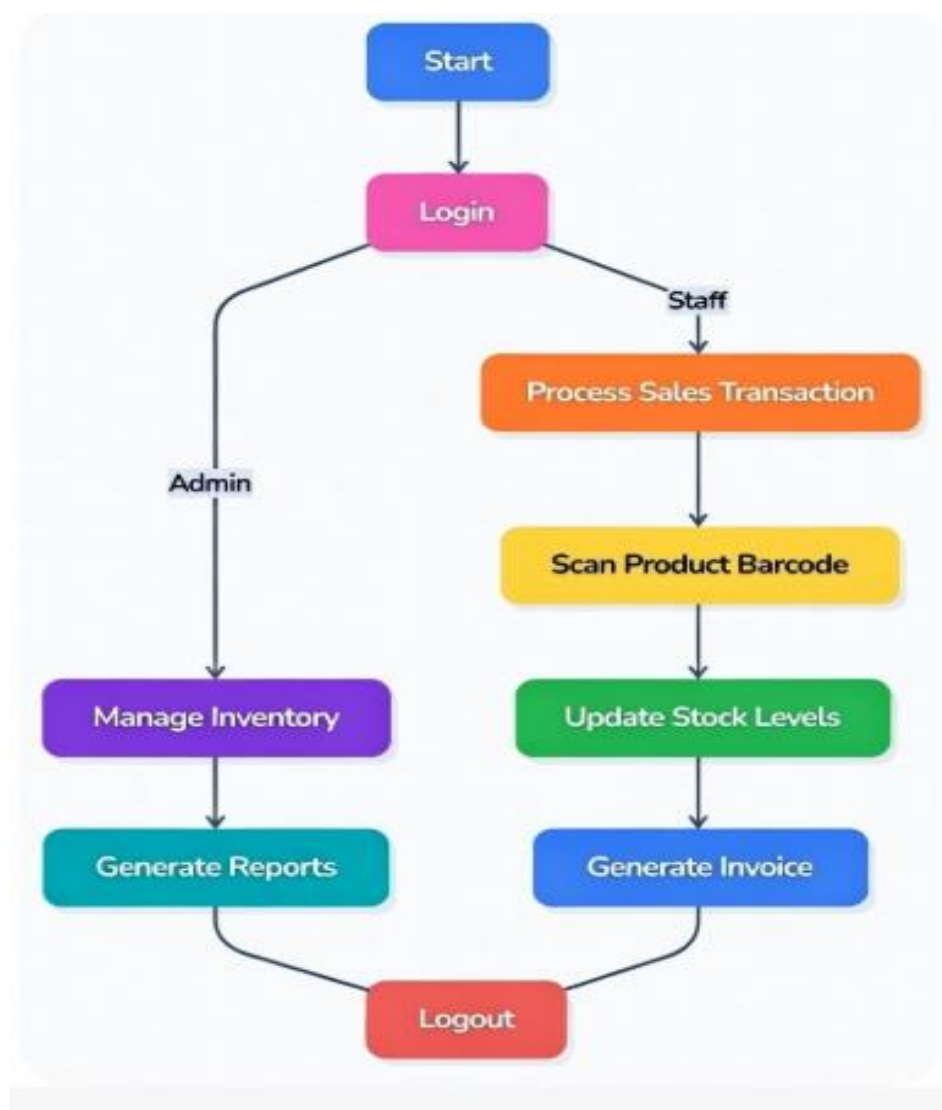
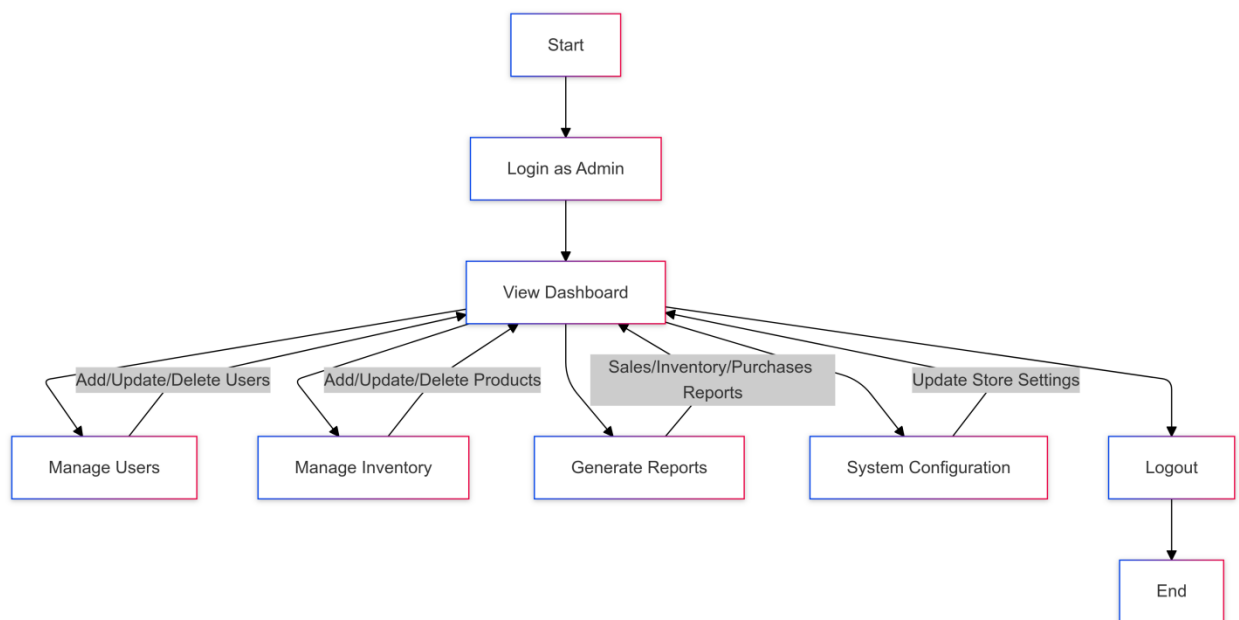


Figure 3.2.3: Activity Diagram

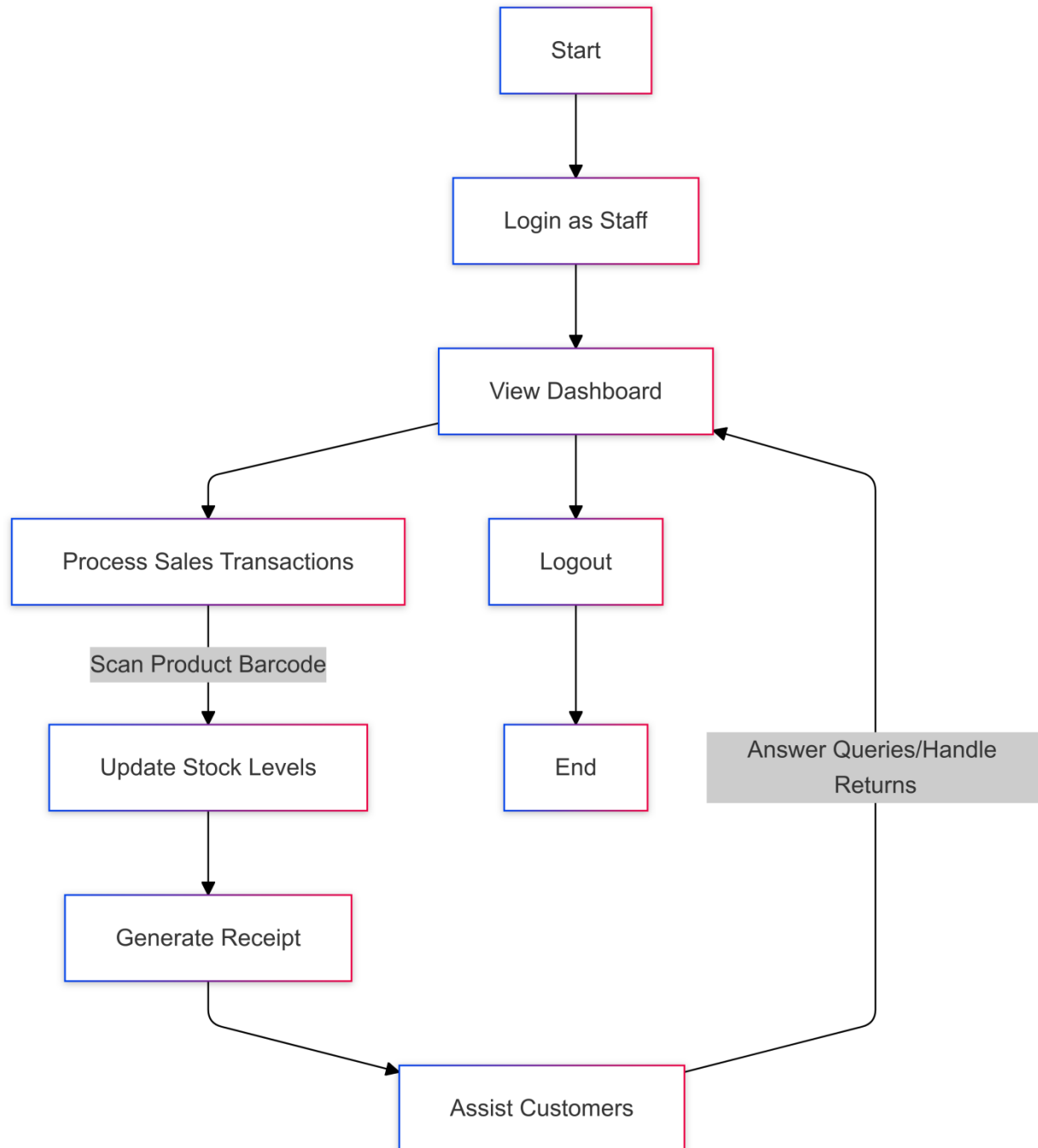
3.2.3 Sequence Diagram

A Sequence Diagram, part of the Unified Modeling Language (UML), is an interaction diagram that models the chronological sequence of messages exchanged among objects, actors, or components during the execution of a specific scenario or use case. It emphasizes the temporal order of interactions, lifelines of participating elements, and the flow of control or data in a dynamic system context.

3.3.3.1 Admin



3.3.3.1 Staff



4. Data Structure

The system uses a strong data layout to keep information safe and organized.

Database Type:

- Relational structure.
- Main tables: Users, Products, Sales, Purchases.

Normalization:

- Uses third normal form to avoid repetition.

Tables:

Users:

user_id, username, password, role, email

Products:

product_id, name, details, price, quantity, category_id

Sales:

sale_id, user_id, date, total

Purchases:

purchase_id, supplier_id, date, cost

Connections:

- One User can have many sales
- Products and sales are linked by another table
- One supplier can have many purchases.

Backup:

- Daily backup system.
- Recovery steps in emergencies

5. USER INTERFACE DESIGN

5.1 Overview of User Interface

The system interface is clear, simple, and friendly. It allows staff and admin to work quickly without confusion.

5.2 Screen Images

5.2.1 Home Screen

Inventory System Admin

2 System Users [More info](#)

5 Category [More info](#)

7 Product [More info](#)

6 Customer [More info](#)

3 Supplier [More info](#)

4 Total Purchase [More info](#)

3 Total Outgoing [More info](#)

List of Products [+ Add Products](#)

Show 10 entries Search:

ID	Name	Price	Qty.	Image	Category	Actions
1	Product One	20	181		Category One	Edit Delete
2	Product Two	50	121		Category Two	Edit Delete
3	Product Three	13	65		Category Three	Edit Delete
4	Product Four	30	65		Category Four	Edit Delete
5	Product Five	39	88		Category Three	Edit Delete
6	Product Six	66	166		Category Four	Edit Delete
7	Test Product	56	79		Category Five	Edit Delete

Showing 1 to 7 of 7 entries [Previous](#) [Next](#)

5.2.2 Login Screen

Logical screen design defines the layout and functionality of each screen in the INVENTORY MANAGEMENT SYSTEM. It outlines how users interact with the system, ensuring clarity and efficiency.

6. Appendices

6.1 References

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